Unit 4 · Outcome 3 · Scientific Investigation

Health, Safety & Ethics

KK05 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

A brilliant method that is unsafe or unethical scores nothing.

U·O3 · KK05 Scientific Investigation A brilliant method that is unsafe or unethical scores nothing. Every student-designed investigation in VCE Environmental Science has to clear two gates before a single measurement counts: a safety gate (no one and nothing gets hurt collecting the data) and an ethics gate (the study is honest, fair, and respectful of people, living things and Country). This module is about identifying — in writing, the way VCAA wants it — exactly which health, safety and ethical guidelines apply to your chosen investigation. SAFE → ETH

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01 Why this dot point exists It is the dot point that protects everything else. The other dot points in this chapter are about getting good data — the right methodology, variables, accuracy and precision. KK05 sits underneath all of them. VCAA states plainly that knowledge of the safety considerations and ethical standards associated with environmental science investigations is integral to the study . In the exam this shows up two ways: Applied to a given scenario — you'll be handed someone's investigation (a creek survey, a soil study, a household-energ

Meet the Wirrim Creek study.

02 The chapter anchor Meet the Wirrim Creek study. We'll carry one running investigation through this whole chapter so the abstract terms have something concrete to stick to. It's a classic, very doable Year 12 fieldwork study — the kind hundreds of VCE students actually run. ⬡ Investigation file · Wirrim Creek Does the Marrin Street stormwater drain affect creek health? Priya , a Year 12 student, investigates whether water quality differs upstream and downstream of a stormwater outfall on Wirrim Creek , in a reserve managed by the local council. She use

First, separate a hazard from a risk.

03 Health & safety First, separate a hazard from a risk. Students throw these words around as synonyms and lose marks for it. They are not the same thing, and exam questions deliberately test the difference. Hazard Risk Risk assessment The phrasing examiners reward "The fast-flowing water is a hazard ; the risk is that the investigator could slip on the bank and drown. This risk is reduced by never sampling alone and staying out of water above knee depth (control)." Hazard → risk → control, every time. Fieldwork hazards hide in the boring details. Enviro

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Not all controls are equal — climb the ladder from the top.

04 Hierarchy of controls Not all controls are equal — climb the ladder from the top. Once you've identified a risk, you control it. The hierarchy of controls ranks control methods from most effective (remove the hazard entirely) to least effective (rely on the person wearing gear). Strong answers reach for the highest practical control, not just "wear gloves". Click a rung to see it applied to Wirrim Creek. ↑ most effective least effective ↓ Reaching for PPE first loses marks PPE is the bottom rung — it protects one person and fails if forgotten. Always

The five ethical concepts — learn the exact words.

05 The ethics gate The five ethical concepts — learn the exact words. VCAA assesses five named ethical concepts across all the sciences. You need to be able to define each and apply it to an investigation. The mnemonic "I Just Brought New Reagents" keeps them in order: I ntegrity · J ustice · B eneficence · N on-maleficence · R espect. Tap a concept to see its definition and how it lands on the Wirrim Creek study. Beneficence vs non-maleficence — the classic mix-up Beneficence = actively maximise the benefits (the study informs council's drain management

What ethics looks like for environmental fieldwork.

06 Ethics in the field What ethics looks like for environmental fieldwork. The five concepts are the framework; here is how they translate into concrete obligations for the kinds of investigations you'll actually run. Living things Handle organisms gently, minimise time out of water, return them to the same spot , and never collect more than needed. In VCE you are not expected to use animals , and may do so only where it complies with the law. This is respect and non-maleficence in action. The environment itself Cause minimal disturbance : stay on tr

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Audit the Wirrim Creek plan.

07 Interactive Audit the Wirrim Creek plan. Here are eleven things from Priya's plan. Sort each into the right bin — is it a safety hazard/control , an ethical concern , or just a method detail that's neither? Drag on desktop, or tap a token then tap a bin on touch. Hit Check when you're done. Safety hazard / control Ethical concern Method detail only Check my sorting

Read the command word before you write.

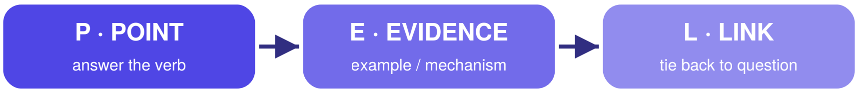
◈ **COMMAND WORDS**

08 Decoder Read the command word before you write. Safety-and-ethics questions live or die on the verb. "Identify" wants a quick label; "justify" wants reasons; "evaluate" wants both sides. Tap a command word to see exactly what it's asking and how to answer it in this topic.

Structure every ethics answer with P·E·L.

✎ P·E·L WRITING

09 Writing trainer Structure every ethics answer with P·E·L. The pattern that turns a vague answer into a full-mark one: Point (name the concept/hazard), Example (tie it to the specific investigation), Link (say how it's addressed or why it matters). Try it below — write a 2-mark answer to the prompt, and the checklist lights up as you hit each part. P · POINT Name it precisely Use the exact term — "non-maleficence", not "being kind to the bugs". E · EXAMPLE Anchor to the study Refer to this investigation — the macroinvertebrates, the drain, the survey.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Six ways students throw away these marks.

10 Exam traps Six ways students throw away these marks. T1 Treating "hazard" and "risk" as the same word A hazard is the source of harm (deep water); the risk is the chance and severity of harm from it (drowning). Fix: write hazard → risk → control as three distinct things. T2 Defaulting to "wear gloves / be careful" PPE and vague care are the weakest controls. Fix: climb the hierarchy — can you eliminate or substitute the hazard, or use an engineering/administrative control first? T3 Everyday words instead of the ethical concepts "Being nice to the envi

Exam-style questions — write, then reveal & self-mark.

★ PRACTICE & SELF-MARK

11 Practice · 20 marks Exam-style questions — write, then reveal & self-mark. Attempt each in your logbook before revealing. Compare against the weak and strong responses, read the marking guide, then award yourself a mark. Your total saves automatically in the dock. Q1 Distinguish · short answer Using the Wirrim Creek investigation, distinguish between a hazard and a risk . 2 marks Reveal responses ✗ Weak response "A hazard and a risk are both dangerous things at the creek, like the water and slipping over." Treats the two terms as synonyms, gives no cl

Cheat sheet.

✦ **RECAP / CHEAT SHEET**

12 One-screen recap Cheat sheet. Term Nail it in the exam as… Hazard A source of potential harm (deep water, a reagent, a snake). Risk The likelihood & severity of harm from a hazard (chance of drowning). Hierarchy of controls E liminate → S ubstitute → I solate → E ngineering → A dministrative → PPE . Aim high. SDS Safety Data Sheet — a chemical's hazards, handling, first aid & disposal. Integrity Honest, accurate recording & reporting of all results, open to scrutiny. Justice Fair consideration of all claims; no group unfairly burdened; fair access to

Term	Nail it in the exam as…
Hazard	A source of potential harm (deep water, a reagent, a snake).
Risk	The likelihood & severity of harm from a hazard (chance of drowning).
Hierarchy of controls	E liminate → S ubstitute → I solate → E ngineering → A dministrative → PPE . Aim high.
SDS	Safety Data Sheet — a chemical's hazards, handling, first aid & disposal.
Integrity	Honest, accurate recording & reporting of all results, open to scrutiny.
Justice	Fair consideration of all claims; no group unfairly burdened; fair access to benefits.
Beneficence	Maximise benefits ; the study should do good.
Non-maleficence	Avoid/minimise harm ; harm not disproportionate to benefit.
Respect	Welfare, autonomy, beliefs, culture & intrinsic value of living things.
Three approaches	Consequences-based · duty/rule-based · virtues-based.
Human participants	Informed consent · voluntary · confidential/anonymous · right to withdraw.
Fieldwork & Country	Minimal disturbance · permits/permission · respect Traditional Owners & heritage.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

A hazard and a risk are both dangerous things at the creek, like the water and slipping over." Treats the two terms as synonyms, gives no clear distinction, and doesn't separate source from consequence. 0–1.

✓ STRONG / MODEL ANSWER

A hazard is a source of potential harm — the fast-flowing, deep water of the creek. The risk is the likelihood and seriousness of harm arising from that hazard — that Priya could slip on the bank, fall in and drown." Defines both, and clearly separates the source (water) from the chance/severity of harm (drowning). 2/2.

✗ WEAK ANSWER

Hazard 1: the water. Hazard 2: snakes. She should wear gloves and be careful." Hazards are vague, and both "controls" are PPE/be-careful — the weakest rung. No hierarchy reasoning. ~1.

✓ STRONG / MODEL ANSWER

Hazard 1 — deep/fast water (drowning). Control: an administrative control — sample only in shallow margins (knee-deep max) with a buddy present, never alone. Hazard 2 — snakes/bites in long grass . Control: substitute the route for a mown track plus PPE (long pants, sturdy boots) — choosing the highest practical control first." Two specific hazards, each with a control explicitly placed in the hierarchy, and a preference for higher controls. 4/4.

✗ WEAK ANSWER

She should respect the bugs and be honest. That's good ethics." Gestures at respect and integrity but never names them as concepts or applies them precisely. ~1.

✓ STRONG / MODEL ANSWER

Non-maleficence — avoiding harm: she returns every macroinvertebrate to the water unharmed and takes no more than needed. Integrity — honest reporting: she records and reports all counts accurately, including downstream data that may weaken her hypothesis, rather than discarding inconvenient results." Two concepts correctly named, each defined briefly and applied to a specific feature of the study. 4/4.

✗ WEAK ANSWER

Be polite and don't share their answers with the class." Touches confidentiality loosely; misses informed consent entirely and gives no real justification. ~1.

✓ STRONG / MODEL ANSWER

Informed consent — participants must be told the survey's purpose and how data will be used, and agree freely; this respects their autonomy and lets them make a real choice (voluntary participation, with the right to withdraw). Confidentiality/anonymity — responses must be de-identified and stored securely, because people share honest data about their household only if their privacy is protected (respect and non-maleficence)." Two correct guidelines, each justified with a reason grounded in an ethical concept. 4/4.

✗ WEAK ANSWER

It's mostly fine. Gloves are good for safety. Maybe she should also wear sunscreen and be nice to the animals." No real evaluation, only one weak addition, everyday language, no ethical concepts or hierarchy. ~1–2.

✓ STRONG / MODEL ANSWER

The statement is partly valid: gloves are a legitimate PPE control (hygiene against waterborne bacteria), and avoiding harm to people relates to non-maleficence . However it is incomplete. (1) Safety beyond PPE — it ignores higher-order controls for drowning (administrative: buddy system, shallow margins only) and other hazards (sun, snakes); PPE is the weakest rung. (2) Animal welfare — non-maleficence & respect — the macroinvertebrates must be returned unharmed and not over-collected. (3) Permission & Country — justice & respect — council access permission and respect for Traditional Owners and cultural heritage are required. Integrity in honest reporting is also unaddressed. Overall the c

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.